

Conditional-variance/forest bridge, stationary-baseline residual, and the root-shell operator boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1. Purpose and scope

R-124 isolated one coefficient-level gate: match the R-121 covariance-normal endpoint, the R-123 primitive trace, and the R-063 forest exactly once before attempting a correlated root-shell estimate. Three independent proof routes were run against that gate. Their common conclusion corrects the proposed three-object equality: conditional future averaging creates a load-bearing variance rebate. Omitting it fails on a deterministic production fixture.

The corrected identity is exact at finite cutoff. It also exposes the exact owner-resolved stationary-baseline residual and a sufficient common-terminal Doob criterion under which the complete low-plus-root aggregate is nonpositive. A root-only upper bound additionally needs a nonnegative complete-low stationary atom. The production middle-current family is root dependent, so neither extra hypothesis nor the required stationary-baseline upper bound is proved here. Individual stationary atoms need not vanish or have one sign.

A separate Hilbert-matrix lemma gives a sharp quantitative acceptance test for the remaining correlated root-shell operator. It uses all root and shell sums inside expectation and requires no separated fifth moment, no $10/3$ moment, and no BMO estimate. The production total-symbol factorisation and its decay are not proved.

R-125 conclusion. At fixed cutoff, positive floor, common legal heat and covariance, and an orthogonal exhaustive contraction-closed output partition, each fixed root owner k , after recombining its same-root visits v , satisfies

$$\Psi_k = \Delta\mathcal{V}_{\text{fut},k} - 2\mathbb{E}_{f,k} \sum_{v \in \mathcal{V}(k)} (\mathcal{R}_{Q,k,v} + \mathcal{M}_{U,k,v} + \mathcal{K}_{R,k,v}) = \Delta\mathcal{V}_{\text{fut},k} - \Delta F_{063,k}^{\text{ad}}.$$

After one owner sum, the complete R-124 secant is

$$\Psi := \sum_k \mathbb{E}\Psi_k = \sum_k \mathbb{E}\Delta\mathcal{V}_{\text{fut},k} - \sum_k \mathbb{E}\Delta F_{063,k}^{\text{ad}}.$$

The variance term is indispensable. The 2680/729 Cartan connection remains inside the covariance-normal endpoint.

At the stationary endpoint the exact residual is

$$B_{J,\pi} := \sum_{k,\mathcal{C}} \mathbb{E}S_{0,k,\mathcal{C}} = \sum_{k,\mathcal{C}} \mathbb{E}\mathcal{V}_{0,k,\mathcal{C}} - \sum_k \mathbb{E}F_{063,0,k}^{\text{ad}},$$

up to the explicit complement leakage when the output partition is not exhaustive. A common-terminal Doob lemma makes the complete low-plus-root aggregate nonpositive; the root-only bound additionally requires a nonnegative complete-low stationary atom. Neither condition is proved for the actual k -dependent production currents. Thus the R-124 stationary-baseline bound remains open.

For a fully recombined far operator with $\|A_{kj}\| \leq C_* 2^{j-4k}$, the Hilbert–Schmidt test gives $K_{\text{far}} = 8C_* 2^{-3j_0} / \sqrt{16065}$. The check $K_{\text{far}} < 4\sqrt{\eta\zeta}$ is sufficient for the R-124 budget. The actual production factorisation, adapted continuum forest bound, balanced band estimate, OVERLAP_{src}, Nelson, removals, the interacting measure, and Sector-A closure remain open. Tier stays T4.

2. Conditional six-row setup

Let $\mathcal{G}$ be the strict past, retain the current root ξ , and put

$$\mathcal{H} = \mathcal{G} \vee \sigma(\xi), \quad \mathbb{E}_f = \mathbb{E}[\cdot \mid \mathcal{H}]. \quad (2.1)$$

Thus $\mathbb{E}_f$ integrates only legal future variables. The heat dummy r remains a coordinate of $L^2(\mu_\Sigma)$; it is not averaged away before the output norm is formed. Let G'_{own} be independent of the past, the current and future roots, and the heat dummy, with the once-owned covariance packet Γ_{own} . Sections 2–4 fix one root owner; its owner index is suppressed and v denotes same-root visits. Across the low and root owners,

$$\Gamma_{\text{target}} = \Gamma_\ell + \sum_k \Delta\Gamma_k. \quad (2.1a)$$

Full covariance means retaining every off-diagonal entry inside each owned packet, not repeating Γ_{target} at every root or visit.

For endpoint $u \in \{0, h\}$, let

$$J_u = C_6(W_{u,r})V_u \quad (2.2)$$

denote the unaveraged full six-row current, with all later feedback already inside W_u, V_u before squaring. Let $\{P_{\mathcal{C}}\}$ be mutually orthogonal contraction-closed output projections satisfying

$$P_{\mathcal{C}}P_{\mathcal{D}} = 0 \quad (\mathcal{C} \neq \mathcal{D}), \quad \sum_{\mathcal{C}} P_{\mathcal{C}} = I. \quad (2.3)$$

Define

$$\Phi_{u,\mathcal{C}} = \mathbb{E}_f[P_{\mathcal{C}}J_u], \quad (2.4)$$

$$\Theta_{u,\mathcal{C}} = \mathbb{E}_{f,G'_{\text{own}}} \|P_{\mathcal{C}}C_6(W_{u,r})\partial G'_{\text{own}}\|^2, \quad (2.5)$$

and the conditional future variance

$$\mathcal{V}_u = \sum_{\mathcal{C}} \mathbb{E}_f \|P_{\mathcal{C}}J_u - \Phi_{u,\mathcal{C}}\|^2. \quad (2.6)$$

The A1 coefficients in the R-120/R-123 convention are

$$c_0 = \frac{3}{250P}, \quad c_1 = \frac{243}{8000P}, \quad \alpha = \frac{5}{9}, \quad P = 4 + 10^{-12}, \quad (2.7)$$

and the fixed six-row frame obeys

$$B = C_6^* C_6. \quad (2.8)$$

Equation (2.8), not a state-dependent rank compression, is the bridge used below.

3. Exact conditional-variance/covariance-normal bridge

Theorem 3.1 (primitive trace and future-variance rebate)

Under (2.1)–(2.8), at each endpoint

$$\sum_{\mathcal{C}} \Theta_{u,\mathcal{C}} = \mathbb{E}_f \sum_i \int (P_{\Sigma}B)(U_u) : (\Gamma_{\text{own}})_i \, dx, \quad (3.1)$$

and

$$\mathbb{E}_f \sum_i \int (P_{\Sigma}B)(U_u) : V_{u,i}^{\otimes 2} \, dx = \sum_{\mathcal{C}} \|\Phi_{u,\mathcal{C}}\|^2 + \mathcal{V}_u. \quad (3.2)$$

Consequently the covariance-normal endpoint $\mathcal{E}_u^{\text{CN}}$ satisfies

$$\boxed{\mathbb{E}_f \mathcal{E}_u^{\text{CN}} = \frac{1}{2} \sum_{\mathcal{C}} (\|\Phi_{u,\mathcal{C}}\|^2 - \Theta_{u,\mathcal{C}}) + \frac{1}{2} \mathcal{V}_u.} \quad (3.3)$$

Proof. The orthogonal resolution (2.3), Parseval, and (2.8) give

$$\sum_{\mathcal{C}} \mathbb{E}_{f,G'_{\text{own}}} \|P_{\mathcal{C}}C_6(W)\partial G'_{\text{own}}\|^2 = \mathbb{E}_f \text{Tr}(C_6(W)\Gamma_{\text{own}}C_6(W)^*) = \mathbb{E}_f B(W) : \Gamma_{\text{own}}.$$

Heat integration and the spatial derivative sum give (3.1). Conditional Pythagoras gives

$$\mathbb{E}_f \|P_{\mathcal{C}}J_u\|^2 = \|\Phi_{u,\mathcal{C}}\|^2 + \mathbb{E}_f \|P_{\mathcal{C}}J_u - \Phi_{u,\mathcal{C}}\|^2.$$

Summing proves (3.2), and subtracting (3.1) from (3.2) proves (3.3). $\square$

For the moving-endpoint secant define

$$\Psi_{\mathcal{C}} = \Delta\Theta_{\mathcal{C}} - \text{Re}\langle\Phi_{h,\mathcal{C}} + \Phi_{0,\mathcal{C}}, \Delta\Phi_{\mathcal{C}}\rangle. \quad (3.4)$$

For the fixed root owner of Sections 2–4, R-121 gives the complete moving-endpoint visit telescope. Taking the difference of (3.3) therefore yields

$$\boxed{\mathbb{E}_f \sum_v (\mathcal{R}_{Q,v} + \mathcal{M}_{U,v} + \mathcal{K}_{R,v}) = \frac{1}{2} \left(\Delta\mathcal{V} - \sum_{\mathcal{C}} \Psi_{\mathcal{C}} \right).} \quad (3.5)$$

Equivalently,

$$\boxed{\sum_{\mathcal{C}} \Psi_{\mathcal{C}} = \Delta\mathcal{V} - 2\mathbb{E}_f \sum_v (\mathcal{R}_{Q,v} + \mathcal{M}_{U,v} + \mathcal{K}_{R,v}).} \quad (3.6)$$

Every intermediate visit endpoint cancels. Thus (3.6) has no visit-count factor and descends through every legal representation-preserving same-root subdivision.

If (2.3) is not exhaustive, (3.5) gains $\Delta\mathcal{L}/2$, where

$$\mathcal{L}_u = \mathbb{E}_f \langle J_u, (I - \sum P_{\mathcal{C}}) J_u \rangle - \mathbb{E}_{f, G'_{\text{own}}} \langle T_u, (I - \sum P_{\mathcal{C}}) T_u \rangle. \quad (3.7)$$

Contraction closure alone does not make (3.7) vanish.

4. Finite-cutoff adapted partial-Wick coordinate

Let Y_{λ} be fresh-root linear coordinates and put

$$Q_{\lambda\mu} = Y_{\lambda} Y_{\mu} - \Gamma_{\lambda\mu}. \quad (4.1)$$

For a smooth cylindrical symmetric coefficient $F_{\lambda\mu}$, directional Malliavin derivatives give the exact identity

$$\boxed{F : Q = \sum_{\lambda,\mu} [F_{\lambda\mu} \diamond_j Q_{\lambda\mu} + Y_{\lambda} D_{\mu} F_{\lambda\mu} + Y_{\mu} D_{\lambda} F_{\lambda\mu} - D_{\lambda} D_{\mu} F_{\lambda\mu}].} \quad (4.2)$$

Indeed, apply $F \diamond_j Y = FY - D_Y F$ twice and use Wick associativity.

For visit v , take $F = B_v - B_{v-1}$ and define

$$F_{RQ,v}^{\text{ad}} = (B_v - B_{v-1}) : Q_{v-1} = 2\mathcal{R}_{Q,v}. \quad (4.3)$$

Then (3.6) becomes the fully owner-resolved finite-cutoff formula

$$\boxed{\sum_{\mathcal{C}} \Psi_{\mathcal{C}} = \Delta\mathcal{V} - \mathbb{E}_f \sum_v F_{RQ,v}^{\text{ad}} - 2\mathbb{E}_f \sum_v (\mathcal{M}_{U,v} + \mathcal{K}_{R,v}).} \quad (4.4)$$

Writing the fixed-root complete covariance-normal endpoint without its outer 1/2 as $F_{063,u}^{\text{ad}}$ gives the compact version

$$\boxed{\Psi = \Delta\mathcal{V} - \Delta F_{063}^{\text{ad}}.} \quad (4.5)$$

For $F = B(Z_*)$, chain rule gives

$$D_j F = DB(Z_*)[D_j Z_*], \quad (4.6)$$

$$D_j^2 F = D^2 B(Z_*)[D_j Z_*, D_j Z_*] + DB(Z_*)[D_j^2 Z_*]. \quad (4.7)$$

Thus adapted Dh, D^2h families are visible explicitly. This proves an algebraic coordinate at finite cutoff; it does not prove their continuum root-shell bounds.

The deterministic R-063 forest is recovered exactly. If Z, Y are jointly Gaussian, $\mathbb{E}Z^2 = \sigma$, $\mathbb{E}Y^2 = \gamma$, $\mathbb{E}ZY = k$, and $Q = Y^2 - \gamma$, (4.2) gives

$$ZQ =: ZYY : + 2kY, \quad (4.8)$$

$$Z^2Q =: ZZYY : + \sigma Q + 4k : ZY : + 2k^2. \quad (4.9)$$

Hence the ΣQ term and every lower contraction remain. When h is deterministic, dyadic Taylor regrouping yields R-063's P_3/P_1 , $P_4/P_2^{\text{cross}}/\Sigma Q/P_0^{\text{cross}}$, and remainder chart. For adapted h , the top Wick factor may have arbitrary chaos order; the registered finite-forest theorem and its hypercontractive estimates do not transfer automatically.

5. Exact counterfixture to the naive identification

Take trivial past, one future $\eta \sim N(0, 1)$, no heat, one full output cluster, base $w_0 = 0$, terminal $w_1 = e_1$, derivative

$$V = \eta e_2, \quad \Gamma = e_2 e_2^T. \quad (5.1)$$

This is a deterministic constant translation, already inside R-063's scope. Only the S_1 pair survives. With

$$s = c_0 + c_1 = \frac{339}{8000P}, \quad (5.2)$$

one has

$$J_1 = 2\eta(\sqrt{c_0}, \sqrt{c_1}), \quad \Phi_1 = 0, \quad \Theta_1 = 4s, \quad \mathcal{V}_1 = 4s. \quad (5.3)$$

The covariance-normal endpoint and deterministic R-063 forest both have mean zero:

$$\mathbb{E}[2s(\eta^2 - 1)] = 0. \quad (5.4)$$

Formula (3.5) reads $0 = (4s - 4s)/2$. The naive equality without conditional variance would instead give $0 = -2s$ and miss exactly

$$2s = \frac{339}{4000P}. \quad (5.5)$$

Therefore the variance rebate is necessary even before adaptation. This fires the naive primitive-trace/forest identification route, not the corrected bridge.

6. Owner-resolved stationary-baseline residual

After every representation-preserving same-root visit has been recombined, the stationary current is a root family $J_{0,k}$, not in general the projections of one common terminal current. Root k uses its once-owned covariance increment $\Delta\Gamma_k$, while the complete-low covariance Γ_ℓ remains a distinct owner:

$$\Gamma_\ell + \sum_k \Delta\Gamma_k = \Gamma_{\text{target}}. \quad (6.1)$$

A same-root subdivision creates no new covariance owner: it keeps $\Delta\Gamma_k$ fixed and sums only moving-endpoint differences. Repeating Γ_{target} at each root, or adding $\Delta\Gamma_k$ as a fresh primitive trace at each same-root subvisit, duplicates the trace. Let

$$\Phi_{0,k,\mathcal{C}} = \mathbb{E}_f[PCJ_{0,k}], \quad \mathcal{V}_{0,k,\mathcal{C}} = \mathbb{E}_f\|PCJ_{0,k} - \Phi_{0,k,\mathcal{C}}\|^2, \quad (6.2)$$

and let $\Theta_{0,k,\mathcal{C}}$ use an independent derivative with covariance $\Delta\Gamma_k$. Define the actual stationary atom by

$$S_{0,k,\mathcal{C}} = \Theta_{0,k,\mathcal{C}} - \|\Phi_{0,k,\mathcal{C}}\|^2. \quad (6.3)$$

Proposition 6.1 (exact production residual)

For an exhaustive orthogonal output partition, Theorem 3.1 applied root by root gives

$$\boxed{S_{0,k,\mathcal{C}} = \mathcal{V}_{0,k,\mathcal{C}} - 2\mathbb{E}_f\mathcal{E}_{0,k,\mathcal{C}}^{\text{CN}}} \quad (6.4)$$

Define the full, visit-recombined, no-outer-half root reconstruction by

$$F_{063,0,k}^{\text{ad}} := 2\mathbb{E}_f\mathcal{E}_{0,k}^{\text{CN}}. \quad (6.4a)$$

Under the exhaustive orthogonal partition this equals $2 \sum_{\mathcal{C}} \mathbb{E}_f \mathcal{E}_{0,k,\mathcal{C}}^{\text{CN}}$. The complete stationary quantity is therefore

$$B_{J,\pi} := \sum_{k,\mathcal{C}} \mathbb{E} S_{0,k,\mathcal{C}} = \sum_{k,\mathcal{C}} \mathbb{E} \mathcal{V}_{0,k,\mathcal{C}} - \sum_k \mathbb{E} \mathcal{F}_{063,0,k}^{\text{ad}}. \quad (6.5)$$

Here $J_{0,k}$ is visit recombined and $\mathcal{E}_{0,k}^{\text{CN}}$ uses the same once-owned $\Delta\Gamma_k$ as $\Theta_{0,k}$. For a nonexhaustive partition, the exact formula is

$$B_{J,\pi} = \sum_{k,\mathcal{C}} \mathbb{E} \mathcal{V}_{0,k,\mathcal{C}} - \sum_k \mathbb{E} \mathcal{F}_{063,0,k}^{\text{ad}} + \sum_k \mathbb{E} \mathcal{L}_{0,k}. \quad (6.5a)$$

Formula (6.5) or (6.5a) identifies the precise residual; it does not determine its sign or give a cutoff-uniform upper bound.

Lemma 6.2 (conditional common-terminal low-plus-root diagnostic)

Assume additionally that there is one stationary terminal current J_0 such that the complete low output and all centered root/cluster outputs form an orthogonal exhaustive Doob decomposition, and that the once-owned low and root traces form the matching covariance partition. Put

$$b_{0,k,\mathcal{C}} = \mathbb{E}[\Phi_{0,k,\mathcal{C}} \mid \mathcal{F}_{k-1}], \quad Y_{0,k,\mathcal{C}} = \Phi_{0,k,\mathcal{C}} - b_{0,k,\mathcal{C}}. \quad (6.6)$$

The two additional energy identities are

$$\mathbb{E} \|J_0\|^2 = \sum_{\mathcal{C}} \mathbb{E} \|Y_{0,\ell,\mathcal{C}}\|^2 + \sum_{k,\mathcal{C}} \mathbb{E} \|Y_{0,k,\mathcal{C}}\|^2, \quad (6.7)$$

$$\mathbb{E} \|J_0\|^2 = \sum_{\mathcal{C}} \mathbb{E} \Theta_{0,\ell,\mathcal{C}} + \sum_{k,\mathcal{C}} \mathbb{E} \Theta_{0,k,\mathcal{C}}. \quad (6.8)$$

Then, with $b_{0,\ell,\mathcal{C}} = 0$,

$$\sum_{\mathcal{C}} \mathbb{E} D_{0,\ell,\mathcal{C}} + \sum_{k,\mathcal{C}} \mathbb{E} D_{0,k,\mathcal{C}} = 0, \quad (6.9)$$

and

$$B_{J,\pi}^{\text{full}} := \sum_{\mathcal{C}} \mathbb{E} S_{0,\ell,\mathcal{C}} + B_{J,\pi} = - \sum_{k,\mathcal{C}} \mathbb{E} \|b_{0,k,\mathcal{C}}\|^2 \leq 0. \quad (6.10)$$

Equivalently,

$$B_{J,\pi} = - \sum_{k,\mathcal{C}} \mathbb{E} \|b_{0,k,\mathcal{C}}\|^2 - \sum_{\mathcal{C}} \mathbb{E} S_{0,\ell,\mathcal{C}}. \quad (6.11)$$

Thus the displayed common-terminal hypotheses alone do not imply the root-only R-124 bound $B_{J,\pi} \leq 0$ or $C_0 = 0$. That corollary additionally requires $\sum_{\mathcal{C}} \mathbb{E} S_{0,\ell,\mathcal{C}} \geq 0$, including the special case of equality. Neither the common-terminal identification nor this complete-low sign is proved for production, so R-124 equation (7.3) remains open.

Under the additional hypotheses of Lemma 6.2, neither atom need vanish or have one sign. The following common-terminal fixture has $S_{0,\ell} = 0$ and is diagnostic only; it is not the production stationary-baseline theorem. Let u_k, v_k be independent $N(0, 1/N)$ variables, $U = \sum u_k$, $V = \sum v_k$, $J = UV$, and reveal the first k pairs. With $\Theta_k = 1/N$,

$$\mathbb{E} \|\Phi_k\|^2 = \frac{k^2}{N^2}, \quad \mathbb{E} \|b_k\|^2 = \frac{(k-1)^2}{N^2}, \quad \mathbb{E} \|Y_k\|^2 = \frac{2k-1}{N^2}. \quad (6.12)$$

Therefore

$$\sum_{k=1}^N D_{0,k} = 0, \text{quad} \sum_{k=1}^N S_{0,k} = - \frac{(N-1)(2N-1)}{6N} \leq 0. \quad (6.13)$$

At $N = 2$, the two S_0 atoms are $1/4$ and $-1/2$; only their diagnostic aggregate is $-1/4$. Positive-part or absolute-value summation destroys this conditional cancellation and therefore cannot prove the production bound.

7. Square-first Clark–Ocone coordinate

At finite cutoff, apply martingale representation only after every visit has been synthesized into the terminal endpoint. If

$$J_u - \Phi_u = \int_0^1 U_u(t) \, dB_t, \quad (7.1)$$

then Itô isometry gives

$$\mathcal{V}_u = \mathbb{E}_f \int_0^1 \|U_u(t)\|_{\text{HS}}^2 \, dt. \quad (7.2)$$

Writing $\delta U = U_h - U_0$, equations (4.4)–(4.5) become

$$\boxed{\Psi = \mathbb{E}_f \int_0^1 \{2 \operatorname{Re} \langle U_0, \delta U \rangle + \|\delta U\|^2\} \, dt - \Delta F_{063}^{\text{ad}}.} \quad (7.3)$$

This preserves the mixed covariance inside expectation. Applying Clark–Ocone visitwise would lose quotient safety and reopen the R-118 opposite-visit obstruction.

8. Correlated root-shell operator criterion

Theorem 8.1 (far-tail Hilbert operator budget)

Suppose that only after complete cluster, owner, variance, and forest recombination, the adverse far term has the form

$$T_{\text{far}} = \mathbb{E} \operatorname{Re} \sum_{j \geq j_0} \sum_{k > j} \langle x_k, A_{kj} y_j \rangle, \quad (8.1)$$

where, pathwise,

$$\sum_k \|x_k\|^2 \leq X, \quad \sum_j \|y_j\|^2 \leq Y, \quad (8.2)$$

and

$$\|A_{kj}\|_{\text{op}} \leq C_* 2^{j-4k}. \quad (8.3)$$

Then

$$\|A\|_{\text{op}} \leq K_{\text{far}} := \frac{8C_*}{\sqrt{16065}} 2^{-3j_0}, \quad (8.4)$$

and, for every $\eta_f > 0$,

$$\boxed{T_{\text{far}} \leq 2\eta_f \mathbb{E}X + \frac{K_{\text{far}}^2}{8\eta_f} \mathbb{E}Y.} \quad (8.5)$$

Proof. The scalar dominating matrix has

$$\begin{aligned} \|A\|_{\text{op}}^2 &\leq \sum_{j \geq j_0} \sum_{k > j} C_*^2 2^{2j-8k} \\ &= \frac{64C_*^2}{16065} 2^{-6j_0}. \end{aligned} \quad (8.6)$$

Equations (8.1)–(8.2), Cauchy in the combined root-shell Hilbert space, and one scalar Young inequality prove (8.5). $\square$

For an actual total operator, the sharp mixed-form criterion is

$$\|A\|_{\text{op}} \leq 4\sqrt{\eta\zeta}. \quad (8.7)$$

Since (8.4) is a Hilbert–Schmidt upper bound, the check $K_{\text{far}} < 4\sqrt{\eta\zeta}$ is sufficient, not necessary. The full R-124 trace budgets give

$$4\sqrt{\frac{9}{20}\frac{3}{20}} = \frac{3\sqrt{3}}{5}. \quad (8.8)$$

At action normalization the corresponding coefficient is $3\sqrt{3}/10$. Retaining the disjoint R-103 debts and the R-124 legal-row source allocation gives the stricter trace threshold

$$4\sqrt{\left(\frac{197}{440} - \frac{3}{125P}\right)\frac{3}{25}} = 0.92093233\dots \quad (8.9)$$

The decay (8.3) is not a theorem for the production symbol. R-093’s 2^{j-4k} law is diagnostic, and R-074 forbids importing it into an unsubtracted coefficient. The missing analytic theorem must establish (8.2)–(8.3) for the complete signed symbol

$$\Delta\mathcal{V} - \Delta F_{063}^{\text{ad}} \quad (8.10)$$

and treat the balanced coefficient-dominant band separately.

9. Route map and evidence summary

The parallel routes now have the following reusable verdicts.

1. **Advanced:** the primitive trace, stopped output, conditional variance, and covariance-normal endpoint obey the exact bridge (3.3)–(3.6).
2. **Failed:** primitive trace and forest cannot be identified after conditional averaging without the variance rebate; (5.1)–(5.5) is an exact production counterfixture.
3. **Advanced:** the smooth cylindrical adapted partial-Wick identity (4.2) is exact at finite cutoff and reduces to R-063 in deterministic scope.
4. **Boundary:** arbitrary adapted chaos, Dh, D^2h , the continuum root-shell bounds, and the fifth $H^{-3/5}$ moment are not supplied by R-063.
5. **Advanced conditionally:** Lemma 6.2 gives a nonpositive low-plus-root aggregate under an additional common-terminal Doob and matching trace-partition hypothesis. A root-only upper bound also requires a nonnegative complete-low stationary atom. The exact production residual is (6.5); its bound remains open.
6. **Advanced conditionally:** Theorem 8.1 supplies an explicit operator threshold with no separated rough moment.
7. **Open:** prove the production square ledgers and total-symbol decay for (8.10), plus a signed balanced-band theorem.

The exact multiplicity ledger is: every R_Q, M_U, K_R owner once; intermediate visit endpoints zero; primitive mean-square, trace, and variance rebate once with their displayed half coefficients; heat internal; appended forest zero; later-feedback crosses inside J before squaring; complete low once; paid and REG Cartan owners unchanged. The R-121 curls remain

$$-\frac{40}{729}, \quad \frac{2720}{729}, \quad \frac{2680}{729}. \quad (9.1)$$

No $+40/729$ companion is introduced.

10. Devil’s-advocate review

1. **Objection: the primitive trace is already the R-063 forest. UPHELD AGAINST THE CLAIM.** Conditional averaging changes current energy into mean-square plus variance. Equation (5.5) measures the exact missing half-variance.
2. **Objection: contraction closure makes the output partition complete. UPHELD AGAINST THE CLAIM.** Orthogonality and exhaustivity are separate hypotheses; otherwise the leakage (3.7) remains.
3. **Objection: (4.2) is the adapted R-063 continuum theorem. UPHELD AGAINST THE CLAIM.** It is finite-cutoff algebra. Arbitrary chaos and Dh, D^2h lack the required uniform model bounds.
4. **Objection: stationarity makes every baseline atom zero. UPHELD AGAINST THE CLAIM.** The $N = 2$ fixture has atoms $1/4$ and $-1/2$. Only the low-plus-root aggregate in the common-terminal diagnostic is automatically nonpositive; no production root-only sign is proved.
5. **Objection: the complete low block may be omitted because it is finite. DISMISSED.** Omitting it breaks the conditional Doob identity and can leave a positive root D_0 residual. This does not establish the production baseline bound.
6. **Objection: the Hilbert–Schmidt constant is the sharp production constant. UPHELD AGAINST THE CLAIM.** It is a sufficient majorant. Sharpness belongs to the actual total operator norm, which is not computed.
7. **Objection: R-093 already proves the decay in (8.3). UPHELD AGAINST THE CLAIM.** Its law is diagnostic. The production variance-minus-forest total symbol has not been factorised.
8. **Objection: R-125 closes controlled-shell one-use, Nelson, or Sector A. UPHELD AS AN OVERCLAIM.** The production correlated operator and balanced band remain open. Tier stays T4.

11. Executable verification

The primary exact executable derives the A1 constants from the manifest, checks output Gram traces, conditional Pythagoras, every factor of two in the bridge, the deterministic production counterfixture, the exact baseline residual, conditional common-terminal low-plus-root criterion, and signed N -root diagnostic fixture, Clark–Ocone checks, the three Cartan curls, the geometric-series constant, and every budget threshold. The non-importing standard-library executable uses a different rational frame, four-atom conditional law, three-visit telescope, finite dyadic sums, direct optimization, and several independent stationary root counts. The integrated verifier reruns both, pins all source and authority hashes, checks the note and PDF, and enforces the negative-result, exploration, claim, roadmap, theorem-map, and no-overclaim surfaces.

12. Result footer

Result ID: A13-CLASSII-CONDITIONAL-VARIANCE-FOREST-BRIDGE-
ROOT-SHELL-OPERATOR-BOUNDARY (R-125).

Precise statement: Theorems 3.1 and 8.1; exact Proposition 6.1;
conditional Lemma 6.2; finite-cutoff adapted partial-Wick identity
(4.2); counterfixture (5.1)–(5.5).

Scope: fixed finite cutoff and positive floor; common legal heat, covariance, reveal field, and orthogonal exhaustive output partition; bounded smooth cylindrical controls for adapted algebra; common-terminal Doob and matching trace-partition hypotheses only for Lemma 6.2, with a nonnegative complete-low atom additionally required for root-only $C0=0$; the production stationary baseline is excluded.

Dependencies: A1; R-063, R-074, R-079, R-093, R-103, R-104, R-118, R-119, R-120, R-121, R-123, R-124.

Evidence grade: ANALYTIC, EXACT, EXECUTED.

Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe
 codes/foundations/a13_classii_conditional_variance_forest_bridge_
 root_shell_operator_boundary_verify.py

Expected output: R-125 integrated PASS; primary and independent child counts, hashes, PDF, exploration, negative, and public surfaces pass.

Falsification gate: any failed Gram/trace/variance identity, factor of two, production counterfixture, exact baseline residual, conditional common-terminal low/Doob criterion, dyadic sum, budget threshold, authority/hash/PDF, or scope check.

Tier before / after: T4 / T4.

No-overclaim statement: no adapted continuum forest bound, production root-shell factorization, production stationary-baseline bound or production $C0=0$, balanced-band bound, OVERLAP_src, Nelson, removals, interacting measure, Sector-A closure, or T5-T7 follows.

Next required action: factor the complete signed total symbol
 Delta V_fut-Delta F063_ad inside expectation, prove its far operator norm below the applicable threshold, close the balanced band, and separately prove the owner-complete production stationary-baseline bound.